

Chloe Posluszny, njp 986

Erik Halldorson, ejh2958

Chenhe Yuan, cy8368

Andrew Nguyen, avn486

Praneel Murali, pm33487

Proposed Idea:

Our proposal is for designing and creating a Natural Language Categorization Classifier. The idea is that we'd take natural language queries, which are similar to what could have previously been used to prompt a Large Language Model (LLM) and run them through a series of different machine learning algorithms to categorize them into different groups. This could be used, for example, to label large natural language datasets for different purposes including training machine learning models or testing LLMs.

Data Sourcing:

For sourcing our data, we'd like to experiment with generating queries using a LLM, but as a backup, we have found a dataset containing 60 thousand stack overflow questions ([Stack Overflow Questions Database - kaggle](#)) which should be similar to LLM queries which are specific to the context of programming. We'd also like to experiment with providing a LLM with a sample of queries to generate generalized category labels which can then be used for categorizing the data. Otherwise, we will come up with categories manually.

Estimated Compute Necessary:

Our plan is to combine the use of two self-supervised machine learning algorithms, SBERT and k-means clustering. We will first use SBERT to vectorize each of the queries and categories, then the vectorized queries will be fed into the k-means clustering algorithm to be assigned to different clusters. Finally, we will check the cosine similarity between
?" each of the clusters and each of the categories to determine which category each cluster belongs to. We could also experiment with prompting an LLM to label each cluster of queries with a category of its choosing. We could then compare semantic similarities between each of the queries in a category to make sure they are all similar to test the system. All of these processes would not take much compute power.